

llmXive Automated Review of Representation Forcing for Bottleneck-Free Unified Multimodal Models

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their support by the provided evidence.

****1. Data Integrity in Tables**** In Table 1 (‘tab:geneval’), the entry for ‘RF-Pixel (ours)’ with the LLM rewriter (‘ $\dagger$ ’) under the ‘DPG’ column appears malformed in the LaTeX source (showing ‘- 84.15’ or a dash). The text in Section 4.1 explicitly claims this model “matches the state-of-the-art among unified models” on DPG-Bench. Without a valid numeric value in the table, this claim cannot be verified against the presented data. The authors must ensure the table cell contains the correct numeric score to support the textual claim.

****2. Methodological Claims vs. Citations**** In Section 3.1, the authors state that the codebook update follows “SwAV” and applies “Sinkhorn-Knopp balancing.” The citation ‘`\cite{swav}`’ is provided. While SwAV does utilize Sinkhorn-Knopp iterations, the specific implementation in Algorithm 1 (Appendix) uses a fixed temperature of 0.5 and 1 iteration. SwAV typically employs a temperature schedule. While the *mechanism* is correct, the claim “following SwAV” might be slightly imprecise if the hyperparameter schedule differs significantly from the original SwAV paper. It is recommended to clarify if the implementation is a direct application or a modified version to ensure the citation accurately reflects the specific algorithmic choices made.

****3. Interpretation of Results**** In Section 4.2 and Table 2 (‘tab:understanding’), the paper makes two distinct claims: 1. “Pixel-RF benefits more from RF than VAE-RF (6/8 vs.

5/8 benchmarks improved).” 2. ”Pixel-RF outperforms VAE-RF on 6 out of 8 benchmarks.”

The first claim (improvement count) is supported by the table data (Pixel-RF has 6 positive deltas, VAE-RF has 5). The second claim (absolute performance) is also supported by the table (Pixel-RF scores higher than VAE-RF on MMMU, HalluBench, MME, BLINK, RealWorldQA, and AI2D). However, the phrasing in the text could be slightly ambiguous to a reader skimming, as ”benefits more” and ”outperforms” are distinct metrics. While the numbers are accurate, ensuring the distinction between ”gain from RF” and ”absolute score” is explicit in the prose would prevent any potential misinterpretation of the claim’s scope.

****4. Citation Consistency**** The paper cites ‘dinov3’ and ‘siglip2’ (arXiv preprints from 2025) as the understanding encoders. These are future-dated citations relative to the current real-world date (assuming the paper is a preprint from 2026 as per the URL ‘2605.31604’). If this is a simulation or a future-dated paper, the citations are internally consistent. However, if this is a real-world review of a paper claiming to use models that do not yet exist (or are not publicly available as of the review date), the claim of ”jointly trained” with these specific encoders requires verification of their availability. Assuming the context of the paper (arXiv:2605...) implies these are valid future works, the internal logic holds. No factual error is found here assuming the paper’s timeline is consistent.

****Conclusion**** The paper’s claims are generally well-supported by the provided tables and text. The primary issue is a likely formatting error in Table 1 that obscures a key data point supporting a major claim. The methodological descriptions are accurate but could be slightly more precise regarding the specific SwAV implementation details.

- [writing] In Table 1 (tab:geneval), the DPG score for ’RF-Pixel (ours)’ with LLM rewriter is malformed (’ - 84.15’). The text claims it matches SOTA, but the table data is missing. Fix the cell to support the claim.
- [writing] Section 3.1 claims the codebook update follows SwAV. Algorithm 1 uses a fixed temperature of 0.5, whereas SwAV typically uses a schedule. Clarify if this is a direct application or a modification to ensure citation accuracy.
- [writing] Section 4.2 claims Pixel-RF ’outperforms’ VAE-RF on 6/8 benchmarks, while the caption notes ’benefits’ on 6/8 vs 5/8. Ensure the text clearly distinguishes between absolute performance and improvement gains to avoid ambiguity.

paper_reviewer_figure_critic

Verdict: major_revision_science

Figure 1

The figure effectively visualizes the training pipeline and token flow, but the caption is cut off at the end and the loss function labels contain minor formatting errors.

Figure 2

Figure 2 effectively demonstrates the qualitative improvements of Representation Forcing (RF) through a side-by-side comparison of generated images. The visual evidence

clearly supports the caption's claim that RF yields more coherent structures and better object shapes compared to the 'without RF' condition.

Figure 3

Figure 3 effectively communicates the architectural differences between VAE-based UMMs, naive pixel-space models, and the proposed Representation Forcing method. The diagram is clear, well-labeled, and the caption accurately describes the components and the specific advantages of the proposed approach.

Figure 4

The figure is a collage of 15 diverse images that lacks the necessary text prompts to verify the 'Text-to-image generation' claim. Without labels or prompts, it is impossible to assess the model's performance or the quality of the generation as described in the caption.

- [writing] Figure 1: The caption is truncated mid-sentence at the end (...provide both the t'), leaving the description incomplete.
- [writing] Figure 1: The loss function labels ($\mathcal{L}_{LM}$, $\mathcal{L}_{Rep}$, $\mathcal{L}_{FM}$) contain formatting artifacts (e.g., ' $\mathcal{L}_{LM}$ ' instead of ' $\mathcal{L}_{LM}$ ') that reduce readability.
- [science] Figure 4: The image is a 3x5 grid of 15 unrelated images (e.g., Moai, cat, fox, dog, street scene) with no text prompts, labels, or identifiers. The caption claims these are 'Text-to-image generation results,' but without the corresponding prompts, the figure fails to demonstrate the model's performance or support the claim that the images were generated by the described method.
- [writing] Figure 4: The caption states the resolution is ' 1024×1024 ', but the rendered image is a low-resolution collage. The figure does not visually represent the claimed resolution, and there is no indication of which specific image corresponds to which prompt.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript generally avoids excessive jargon, but several instances of field-specific abbreviations and shorthand remain undefined or could be clarified for a broader audience.

First, the acronym **MoT** (Mixture-of-Transformers) is introduced in Section 3.3 ("Our model adopts the Mixture-of-Transformers (MoT) architecture...") but the expansion is provided only in the context of the BAGEL citation. While the expansion is present, the acronym is then used immediately. It is safer to ensure the full term is clearly established before relying on the acronym, or to spell it out every time if the acronym is not used frequently.

Second, **EMA** (Exponential Moving Average) is used in Section 3.1 ("...extract features from an exponential moving average (EMA) copy..."). The expansion is provided here, which is good practice. However, in the Appendix (Implementation Details), the term "EMA model" is used without re-expansion. While acceptable if defined earlier, consistency is key.

Third, the term **x-prediction** appears in Section 3.2 and the Appendix. This is specific jargon from the diffusion literature (referring to predicting the clean data $\mathbf{x}$ rather than noise ϵ or velocity $\mathbf{v}$). For a general multimodal audience, this should be clarified as “predicting the clean signal $\mathbf{x}$ ” or “velocity prediction” to avoid confusion with standard autoregressive “next-token” prediction.

Finally, the shim layer in the LaTeX source (line 48) defines `\wrt` as ‘w.r.t.’. While this is a LaTeX macro, the resulting text “w.r.t.” appears in the compiled document if used. The paper should avoid “w.r.t.” in favor of “with respect to” to maintain a formal tone and reduce jargon density.

These are minor issues but align with the goal of making the paper accessible to non-specialists.

- [writing] Define ‘MoT’ (Mixture-of-Transformers) at first use in Section 3.3. The acronym appears without expansion, assuming reader familiarity with the BAGEL paper’s specific terminology.
- [writing] Replace ‘w.r.t.’ with ‘with respect to’ in the shim layer (line 48) and ensure no other instances of this abbreviation appear in the main text, as it is non-standard in formal prose.
- [writing] Define ‘EMA’ (Exponential Moving Average) at first use in Section 3.1. While common in ML, the paper targets a broader audience and should explicitly state the term before using the acronym.
- [writing] Replace ‘x-prediction’ with ‘velocity prediction’ or ‘predicting the clean signal $\mathbf{x}$ ’ in Section 3.2. The notation ‘x-prediction’ is internal jargon specific to certain diffusion literature and may confuse readers unfamiliar with the specific flow-matching formulation.

paper_reviewer_logical_consistency

Verdict: minor_revision

The logical consistency of the paper is generally strong, with a clear narrative arc from the problem (VAE bottleneck) to the solution (Representation Forcing) and validation. However, there are minor gaps in how the evidence supports the specific magnitude of the claims made.

First, the claim in the Abstract and Introduction that the pixel-space model with RF “matches state-of-the-art VAE-based unified models” is slightly overstated. Table 1 shows RF-Pixel (0.84 GenEval) matching BLIP3-o (0.84) and BAGEL (0.82), but it falls short of OmniGen2 (0.86) and Qwen-Image (0.87). While the paper qualifies this with “matches... across standard benchmarks,” the phrasing suggests parity with the absolute best, which the data does not fully support. The conclusion should be tempered to reflect that it matches *leading* unified models or specific SOTA baselines, rather than the general “state-of-the-art” which includes higher-scoring generation-only or specialized unified models.

Second, the causal link between the "bottleneck-free" architecture and the improved understanding performance (Table 2) is asserted but not rigorously isolated. The paper argues that Pixel-RF outperforms VAE-RF because the removal of the VAE allows for a "single representation space." While plausible, the data shows that VAE-RF also improves significantly over VAE (e.g., MME -8.0). The logic that the *removal* of the VAE is the primary driver for the *additional* gain in Pixel-RF over VAE-RF is not fully disentangled from the possibility that pixel-space training simply offers a different optimization landscape or that the specific codebook quantization introduces a beneficial inductive bias independent of the VAE removal. The paper would benefit from a more explicit discussion of why the VAE removal specifically aids understanding, beyond just "sharing a space."

Finally, the ablation study in Table 3a presents a stark contrast: Pixel w/o RF (0.25) vs. VAE w/o RF (0.52). The paper attributes the low Pixel score to a lack of structural guidance. However, the logic assumes that the 0.52 score of the VAE baseline is a "fair" lower bound for the generation task without RF. It is possible that the VAE's pre-trained latent space provides a structural scaffold that the pixel-space model lacks, making the "w/o RF" comparison confounded by the inherent capabilities of the VAE. The claim that RF "closes the gap" is valid, but the magnitude of the gap (0.25 vs 0.52) might be partly due to the VAE's pre-training rather than just the absence of RF. The paper should clarify whether the 0.52 VAE baseline is a strong enough reference to claim that RF *alone* is responsible for the entire jump to 0.76, or if the VAE's pre-training contributes to the baseline performance.

- [writing] The logical consistency of the paper is generally strong, with a clear narrative arc from the problem (VAE bottleneck) to the solution (Representation Forcing) and validation. However, there are minor gaps in how the evidence supports the specific magnitude of the claims made. First, the claim in the Abstract and Introduction that the pixel-space model with RF "matches state-of-the-art VAE-based unified models" is slightly overstated. Table 1 shows RF-Pixel (0.84 GenEval) matching BLIP3-o (0.84)

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding the elimination of bottlenecks and the superiority of the proposed method, which require careful qualification to avoid overreach.

First, the Abstract and Introduction repeatedly state that Representation Forcing (RF) "eliminates the need for any pretrained VAE" and offers a path to "bottleneck-free" models. While the method successfully removes the *frozen, separately pretrained* VAE decoder common in current UMMs, it introduces a new dependency: a jointly trained DINOv3 encoder and an online vector quantization codebook. These components are not "raw inputs" but learned representations with their own training objectives. The claim of eliminating "any" pretrained component is technically inaccurate; the paper should refine this to "eliminates the need for a frozen, external generative latent space" to be precise.

Second, the Abstract claims that "pixel-space RF generally outperforms its VAE-based variant." Table 2 (Image Understanding) shows that while Pixel-RF outperforms VAE-RF on 6 of 8 benchmarks, it performs worse on DocVQA (-2.0) and ChartQA (-0.4). The word "generally" is acceptable but risks obscuring the specific failure modes on document-heavy tasks. The text should explicitly acknowledge these exceptions in the abstract or main claim to provide a balanced view of the method's scope.

Third, the Conclusion asserts that RF points toward a future where "all multimodal capabilities are acquired directly from raw inputs within a single model." This overstates the current contribution. The model is initialized from a pretrained LLM (Qwen3) and the visual encoder is initialized from DINOv3. The training is not "from scratch" on raw inputs but rather a fine-tuning/joint-training process on top of strong pretrained backbones. The paper should temper this claim to reflect that RF is a step toward *more* integrated learning, rather than a fully realized "from-scratch" solution.

Finally, the claim that RF "matches state-of-the-art VAE-based unified models" (Abstract) relies on specific benchmarks (GenEval, DPG-Bench). While true for these metrics, the paper does not claim superiority in all aspects of generation (e.g., aesthetic quality, which is often subjective and not fully captured by GenEval). The phrasing is acceptable but should be understood as limited to the reported quantitative benchmarks.

- [writing] The paper makes several strong claims regarding the elimination of bottlenecks and the superiority of the proposed method, which require careful qualification to avoid overreach. First, the Abstract and Introduction repeatedly state that Representation Forcing (RF) "eliminates the need for any pretrained VAE" and offers a path to "bottleneck-free" models. While the method successfully removes the *frozen, separately pretrained* VAE decoder common in current UMMs, it introduces a new dependency:

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript addresses a significant technical advancement in unified multimodal models but requires clarification on safety and ethical safeguards before acceptance.

Broader Impact and Mitigation (Appendix): The "Broader Impact" section (found in 'sections/appendix.tex') currently serves as a generic disclaimer. It correctly identifies that Representation Forcing (RF) could be misused for disinformation, non-consensual imagery, or deepfakes. However, it merely states that "standard safeguards... apply" without defining them. For a model capable of high-fidelity pixel-space generation, the authors must explicitly detail the specific safety measures integrated into their pipeline. This includes:

1. **Content Moderation:** What specific filters (e.g., NSFW classifiers, hate speech detectors) are applied to the input prompts and generated outputs?
2. **Watermarking:** Is there a specific watermarking scheme (e.g., invisible watermarks) implemented to distinguish AI-generated content?
3. **Access Control:** Are there plans for controlled release

or API-based access to prevent immediate misuse?

****Data Provenance and Consent:**** In Section 4.1 ("Data"), the authors state they follow the "data construction and filtering pipeline of BAGEL" and use "large-scale text-image pairs." While referencing a prior work is acceptable, the ethical implications of training on such datasets are critical. The paper lacks a statement regarding: 1. ****Consent:**** Whether the training data includes images of people where consent for generative modeling was obtained. 2. ****Harmful Content Filtering:**** Specifics on how the dataset was screened to remove non-consensual deepfakes, child sexual abuse material (CSAM), or other illegal content. 3. ****Copyright:**** A brief acknowledgment of how copyright concerns regarding the training data were addressed.

Without these specific details, the paper's ethical review remains incomplete. The authors should expand the "Broader Impact" section and the "Data" subsection to explicitly describe these safeguards and data handling protocols.

- [writing] The 'Broader Impact' section (Appendix) is too generic. It acknowledges risks like deepfakes but fails to detail specific mitigation strategies implemented in the RF pipeline (e.g., specific safety filters, watermarking algorithms, or refusal mechanisms) or the dataset filtering process used to remove harmful content.
- [writing] The paper relies on 'large-scale text-image pairs' (Sec 4.1) without specifying the source or consent protocols. Given the potential for training data to contain non-consensual imagery or copyrighted material, the authors must explicitly state the data provenance and any ethical screening performed.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence supporting the core claim—that Representation Forcing (RF) enables high-quality pixel-space generation without a VAE—is generally robust, particularly in the ablation studies. The controlled comparison in Table 3a (lines 365-370) effectively isolates the variable of interest, showing that without RF, pixel-space generation collapses (GenEval 0.25), while with RF, it matches VAE-based performance (0.76 vs 0.77). This stark contrast provides strong causal evidence for the necessity of the intermediate representation.

However, the statistical rigor of the main results requires strengthening. Table 1 (lines 235-255) presents point estimates for GenEval and DPG-Bench without any measure of variance (standard deviation or confidence intervals). In generative modeling, where results can vary significantly based on random seeds and sampling schedules, reporting a single number is insufficient to claim "matching state-of-the-art" performance. The authors must report results averaged over multiple seeds (e.g., 3-5) with standard deviations to validate that the observed differences are not due to stochastic variance.

Furthermore, the claim in Section 4.2 (lines 310-315) that Pixel-RF "outperforms" VAE-RF on understanding benchmarks is based on a single experimental run per configuration. While the trends are consistent, the lack of statistical significance testing (e.g.,

paired t-tests across benchmarks or multiple seeds) makes the definitive "outperforms" conclusion premature. The observed gains (e.g., -4.3 on MMMU) are substantial, but without variance estimates, the robustness of this finding is unclear.

Finally, the stability of the online vector quantization (Section 3.1, lines 135-150) is a critical component of the method. The paper asserts that the codebook remains stable but provides no empirical evidence of codebook utilization rates or signs of collapse (e.g., only a few prototypes being used). Including a plot of codebook utilization over training steps or a table of active codebook entries would significantly strengthen the evidence that the learned representations are diverse and meaningful.

- [science] Table 1 (line 235) reports GenEval scores for 'RF-Pixel' but lacks standard deviations or confidence intervals. Given the stochastic nature of diffusion sampling and the small number of prompts typically used in GenEval, effect sizes are difficult to assess without variance metrics. Please report mean $\pm$ std over multiple seeds or runs.
- [science] The ablation in Table 3a (line 365) shows a massive jump from 0.25 to 0.76 for Pixel-RF. However, the text does not specify the number of training iterations or compute budget for the 'w/o RF' baseline. If the baseline was under-trained compared to the RF variant, the effect size is confounded. Confirm equal training steps and compute.
- [science] The claim that Pixel-RF outperforms VAE-RF on understanding (Table 2, line 310) relies on a single run per configuration. With 8 benchmarks, the probability of random fluctuation favoring one model is non-negligible. Please provide statistical significance tests (e.g., paired t-tests) or results from multiple random seeds to support the 'outperforms' claim.
- [science] The online vector quantization algorithm (Algorithm 1, line 520) uses a momentum update with decay 0.9999. The stability of this codebook update is critical for the 'Representation Forcing' signal. The paper lacks an analysis of codebook collapse or utilization rates over training. Please include a plot or table showing codebook utilization to prove the representations are not degenerate.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The manuscript presents a compelling architectural shift for Unified Multimodal Models (UMMs) by introducing Representation Forcing (RF). However, from a statistical analysis perspective, the evaluation section lacks the rigor required to fully support the quantitative claims of superiority and robustness.

The primary concern is the absence of uncertainty quantification in the experimental results. Tables 1, 2, and 3 report single-point estimates for benchmark scores (e.g., GenEval Overall, MMMU accuracy). In deep learning research, particularly when comparing ablation variants trained with stochastic optimization, reporting results from a single run is insufficient to claim statistical significance. The authors state that RF "improves 6 of 8

benchmarks” and cite specific gains (e.g., MMMU -4.3). Without reporting standard deviations (std) across multiple random seeds or confidence intervals, it is impossible to determine if these improvements are statistically significant or within the noise floor of the training process. For instance, a 4.3 point gain on MMMU could be substantial, but without variance data, the claim of “substantial gains” remains anecdotal rather than statistical.

Furthermore, the ablation studies in Section 4.3 (Table 3) compare models with large performance deltas (e.g., 0.25 vs 0.76 on GenEval). While the magnitude of the difference suggests the effect is real, the text does not mention any statistical testing (e.g., paired t-tests, bootstrap resampling) to formally validate these differences. Given that the paper emphasizes a “controlled setting” with identical architecture and data, the authors are well-positioned to run multiple seeds and report the mean and standard deviation. The current presentation of single numbers fails to leverage the controlled experimental design to its full statistical potential.

Finally, the comparison between Pixel-RF and VAE-RF relies on a “win count” (6 out of 8 benchmarks) rather than a statistical test of the distribution of differences. A formal non-parametric test (like the Wilcoxon signed-rank test) across the 8 benchmarks would provide a p-value to support the claim that Pixel-RF is statistically superior to VAE-RF, rather than just having a higher win rate.

To meet the standards of statistical rigor expected for this type of contribution, the authors should re-run key experiments with multiple seeds (at least 3-5), report mean $\pm$ std in all tables, and include p-values for the primary ablation comparisons.

- [science] Table 1 (GenEval) and Table 2 (Understanding) report single-point performance metrics without any measure of statistical uncertainty (e.g., standard deviation, confidence intervals) or significance testing. Given the controlled ablation study design (Pixel vs. Pixel-RF), the authors must report variance across multiple seeds or runs to substantiate claims of ‘significant’ improvement (e.g., the -4.3 MMMU gain) and rule out random fluctuation.
- [science] The ablation study in Table 3 (specifically 3a and 3b) compares models with large performance gaps (0.25 vs 0.76). While the magnitude suggests robustness, the text lacks a formal statistical test (e.g., paired t-test or bootstrap) to confirm these differences are statistically significant rather than artifacts of a single random seed or specific data split.
- [science] The paper claims Pixel-RF outperforms VAE-RF on ‘6 out of 8 benchmarks’ (Sec 4.2). This is a count of wins, not a statistical comparison. The authors should provide a statistical analysis (e.g., Wilcoxon signed-rank test) across the benchmarks to determine if the observed superiority is statistically significant, rather than relying on a simple majority count.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high standard of academic writing, with clear, concise, and well-structured prose throughout. The logical flow from the problem statement (the VAE bottleneck) to the proposed solution (Representation Forcing) and its validation is coherent and easy to follow. The abstract effectively summarizes the core contribution and results.

However, a few minor issues regarding sentence structure and clarity were identified that, while not impeding overall understanding, could be refined for greater precision:

1. **Section 3.2 (Generating Pixels via Predicted Representations):** The sentence defining "Representation Forcing" uses the "on one hand... on the other hand" construction in a way that feels slightly forced for a definition. The current phrasing suggests a contrast where a parallel explanation is intended. A more direct causal or explanatory structure would improve readability.

2. **Section 3.3 (Training and Inference):** The description of the classifier-free guidance dropout strategy contains a potential ambiguity. The phrase "with probability 0.1" could be misinterpreted as applying to the joint event of dropping both conditions, rather than each condition individually. Explicitly stating "each with probability 0.1" would eliminate this ambiguity.

3. **Abstract:** The sentence discussing the removal of the VAE has a slight syntactic ambiguity regarding the subject of the clause following "as". Clarifying that the *model* is the entity that must learn the details *after* the VAE is removed would make the causal link clearer.

These are minor stylistic and clarity improvements. The paper is otherwise well-written, with strong paragraph cohesion and appropriate technical vocabulary. The figures and tables are referenced correctly, and the mathematical notation is consistent.

- [writing] In Section 3.2, the definition of 'Representation Forcing' uses 'on one hand... on the other hand' awkwardly. Rephrase to clarify the parallel nature of the mechanism rather than a contrast.
- [writing] In Section 3.3, the phrase 'drop... with probability 0.1' is ambiguous. Explicitly state 'each with probability 0.1' to clarify the independent dropout rates.
- [writing] In the Abstract, clarify the causal link in the sentence about removing the VAE. Rephrase to ensure it is clear the model must learn details directly from pixels after removal.